

LAB 03

Title: Checking Whether an Input String is a Single-Line Comment, Multi-Line Comment, or Not a Comment Using C++ with Debugging Techniques

Objective

To learn how to debug a C++ program using Breakpoints and the Watch Window, and to determine whether an input string is a single-line comment, a multi-line comment, or not a comment.

Procedure

1. Complete environment setup.
2. Write the comment detection code.
3. Set break points.

Breakpoint Location	Reason
getline(cin, str);	To verify that the complete input string is read correctly.
if(str.substr(0,2) == "//")	To check whether the input is a single-line comment.
else if(str.substr(0,2) == "/*" && str.substr(str.length()-2) == "*/")	To verify whether the input is a multi-line comment.
else	To identify that the input is not a valid comment.

4. Execute with Next Line.
5. Track the changes on Watch.

Code with break-points

```

CommentChecker.cpp
1  #include <iostream>
2  #include <string>
3  using namespace std;
4
5  int main()
6  {
7      string str;
8
9      cout << "Enter a string: ";
10     getline(cin, str);
11
12     if(str.length() ≥ 2 && str.substr(0,2) == "//")
13     {
14         cout << "Single Line Comment";
15     }
16     else if(str.length() ≥ 4 &&
17             str.substr(0,2) == "/*" &&
18             str.substr(str.length()-2) == "*/")
19     {
20         cout << "Multi Line Comment";

```

```
21 |     }  
22 |     else  
23 |     {  
24 |         cout << "Not a Comment";  
25 |     }  
26 |  
27 |     return 0;  
28 | }
```

Output

